Assignment 3

Course/Topic	Course Instructors
CS 2006 Operating systems Topic Threading Total Marks = 100	Ms. Mahzaib Younas Ms. Sahar Ajmal Ms. Nasreen Akthar Mr. Shahbaz Ayyaz

Assignment Guidelines:

1. Deadlines should be kept in mind. No extension in assignment dates would be given. No late submission will be accepted.
2. This is an individual assignment. **PLAGIARISM IS NOT ACCEPTABLE AT ALL!** Zero marks will be given in case of plagiarism.
3. Make a word document with the convention "SECTION_ROLLNO _LAB-NO".
4. In addition, paste all of your work done at the LINUX prompt. 2.
5. You have to submit a Word File and .cpp files.
6. A single violation of guidelines will lead to Zero marks in your assignment.

Question No 01: **[15]**

Write a multithreaded program that calculates various statistical values for a list of numbers. This program will be passed a series of numbers on the command line and will then create three separate threads. One thread will determine the average of the numbers, the second will determine the maximum value, and the third will determine the minimum value.

For example, suppose your program is passed the integers:

12 75 30 91 49 52 76

The program will report :
The average value is 55
The maximum value is 91
The minimum value is 12
The variables representing the average, maximum, and minimum values will be stored globally.

Question No 02: **[10]**

Create a parent-child relationship between two processes. The parent should print two statements:

A) Parent (P) is having ID <PID>
B) ID of P's Child is <PID_of_Child> The child should print two statements:
C) Child is having ID <PID>
D) My Parent ID is <PID_of_Parent>

Make use of wait() in such a manner that the order of the four statements A, B, C and D is:

A
C
D
B

Question No 03: **[15]**

Design a file-copying program named filecopy using named pipes. This program will be passed two parameters: the name of the file to be copied and the name of the copied file. The program will then

create a named pipe and write the contents of the file to be copied to the pipe. The child process will read this file from the pipe and write it to the destination file.

For example, if we invoke the program as follows: filecopy input.txt copy.txt

The file input.txt will be written to the pipe.

The child process will read the contents of this file and write it to the destination file copy.txt.

Question No 04: [05]

Consider a multiprocessor system and a multithreaded program written using the many-to-many threading model. Let the number of user-level threads in the program be more than the number of processors in the system. Discuss the performance implications of the following scenarios:

a. **The number of kernel threads allocated to the program is less than the number of processors.**
b. **The number of kernel threads allocated to the program is greater than the number of processors but less than the number of user-level threads.**

Question No 05: [05]

Determine if the following problems exhibit task or data parallelism: [also provide proper reason]

- Using a separate thread to generate a thumbnail for each photo in a collection
- Transposing a matrix in parallel
- A networked application where one thread reads from the network and another writes to the network

Question No 06: [05]

Consider the following code segment:

```
pid t pid;
pid = fork();
if (pid == 0) { /* child process */
fork();
thread create( . . .);
}
fork();
```

a. How many unique processes are created?

b. How many unique threads are created?

Question No 07: [15]

The Collatz conjecture concerns what happens when we take any positive integer n and apply the following algorithm:

n = n/2, if n is even

n =3× n + 1, if n is odd

For example, if n = 35, the sequence is 35, 106, 53, 160, 80, 40, 20, 10, 5, 16, 8, 4, 2, 1

Write a C program using the fork() system call that generates this sequence in the child process. The starting number will be provided from the command line. For example, if 8 is passed as a parameter on the command line, the child process will output 8, 4, 2, 1. Since the parent and child processes have their own copies of the data, it will be necessary for the child to output the sequence. Have the parent invoke the wait() call to wait for the child process to complete before exiting the program. Perform necessary

error checking to ensure that a positive integer is passed on the output terminal. You must generate the sequence for more than 2 positive integers.

Question No 08: **[10]**

Implement a code that take two matrix name MATA of size n*m and second Matrix MATB of size m*k and generate a resultant matrix MATC of size n*k by performing multiplication using threads. The number of threads created in system will be equal to n. each thread perform multiplication of each row in MATA with MATB to produced the resultant value for particular row in MATC .

Question No 09: **[10]**

The program shown uses the Pthreads API. What would be the output from the program at LINE C and LINE P?

```
#include <pthread.h>
#include <stdio.h>

int value = 0;
void *runner(void *param); /* the thread */

int main(int argc, char *argv[])
{
pid_t pid;
pthread_t tid;
pthread_attr_t attr;

   pid = fork();

   if (pid == 0) { /* child process */
      pthread_attr_init(&attr);
      pthread_create(&tid,&attr,runner,NULL);
      pthread_join(tid,NULL);
      printf("CHILD: value = %d",value); /* LINE C */
   }
   else if (pid > 0) { /* parent process */
      wait(NULL);
      printf("PARENT: value = %d",value); /* LINE P */
   }
}

void *runner(void *param) {
   value = 5;
   pthread_exit(0);
}
```

Question No 10: **[10]**

Design a multi-threaded program that performs the summation of a non-negative integer in a separate thread using the summation function:

$$sum = \sum_{i=0}^{N} i$$

For example, if N were 5, this function would represent the summation of integers from 0 to 5, which is 15.